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Generating Board Games with MCTS and LLM Agents

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From a plain-text game idea to a playable, tested board game

LLM agents write and revise the game as Python code, MCTS agents play it, and the results of play decide what to change next.

INPUT a game idea in plain text → **LOOP** generate → test → mutate → **OUTPUT** the best playable game found, in Python

Approach: generate, test, mutate, repeat

GAME IDEA
"6×6 board. Pieces are captured when surrounded. Most pieces wins."

plain-text description

1 · GENERATOR

turns the idea into checked game code

- Rule formatter ⇄ Rule validator (retry)
- Planner ⇄ Plan validator (retry)
- Coder ⇄ Code checker (retry)

solid: LLM agent · dashed: deterministic check

{ } game.py

2 · TESTER

MCTS (UCT) self-play

Two identical MCTS agents play 40 games of the new game.

COMPLETION · BALANCE · AGENCY · COVERAGE

fitness = harmonic mean

Kept only if clearly better than the best game so far.

3 · MUTATOR

LLM mutator

Compares the metrics with classic reference games (Checkers, Othello, Connect Four, Nine Men's Morris).

Suggests rule changes in plain text, aimed at the weakest metric.

suggested changes go back to the generator: a new version of the game

Runtime errors go back to the coder. The loop stops after 2 rejected versions in a row or 10 cycles. Evolutionary game design, now with LLMs.

Experimental setup

4 GAMES

- **C1** Connection game with bridge tiles
- **C2** Capture by surrounding, 6×6 board
- **C3** Asymmetric pursuit: King vs 4 Guards
- **C4** Promotion game + a mechanic the LLM invents

MODELS (OPENROUTER, TEMPERATURE 0.7)

GPT-5 · Claude-Sonnet-5
Qwen3.6-Plus · DeepSeek-V4-Pro

PROTOCOL

40 MCTS self-play games per candidate (1 s/move, 60-turn cap). One run per model–game pair: 4 LLMs × 4 games = 16 runs.

METRICS → FITNESS

- **COMPLETION** Share of games that end with a winner
- **BALANCE** How evenly the two players win
- **AGENCY** How often a player has a real choice
- **COVERAGE** How much of the board gets used

All metrics range from 0 to 1. Fitness is their harmonic mean, so one broken metric pulls the score down.

WHEN IS A NEW GAME KEPT?

Only if it is clearly better than the best so far (closes 4% of the gap to 1.0, gains 50%, or makes balance defined) and does not lose on both completion and balance.

REFERENCE GAMES (1,000 MCTS MATCHES)

	Compl.	Balance	Agency	Cover.	Fitness
Checkers	0.363	0.997	0.824	0.499	0.573
Connect Four	0.954	0.977	0.981	0.741	0.901
Othello	0.968	0.994	0.966	0.998	0.981
Nine Men's Morris	0.699	0.933	0.998	0.986	0.885

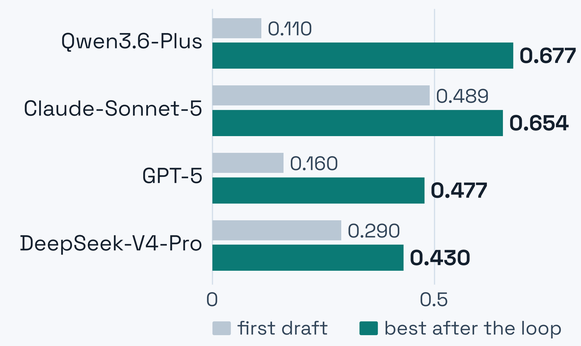
Classic games give the critic reference values. The critic only changes the generated games.

Results

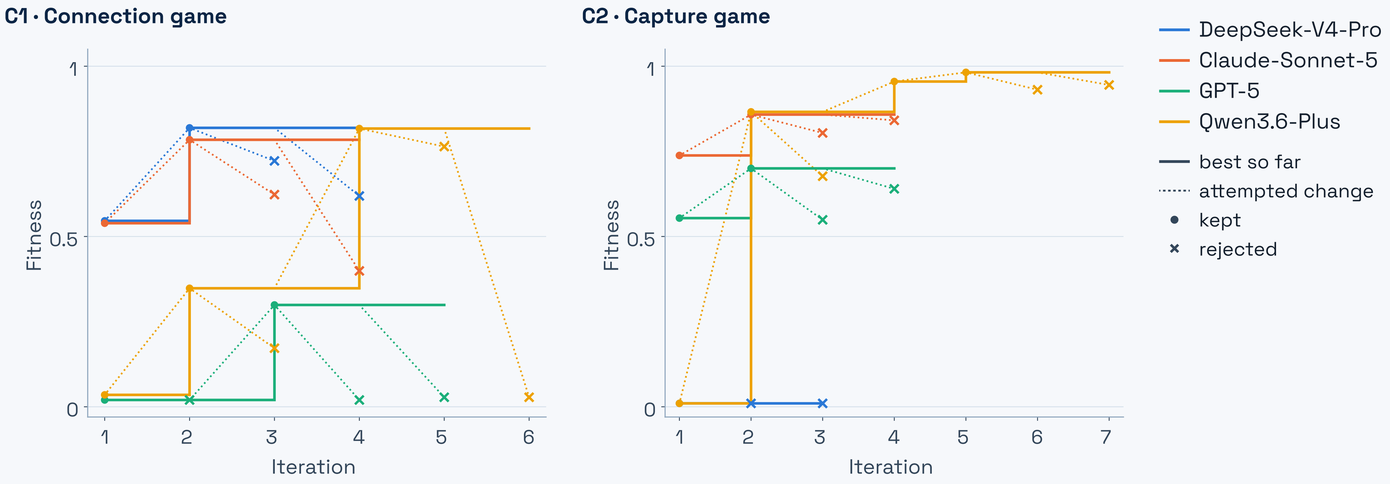
FINAL FITNESS · GAME × MODEL

	DeepSeek V4-Pro	Claude Sonnet-5	GPT-5	Qwen3.6 Plus	Mean
C1 connection	0.819	0.784	0.299	0.817	0.680
C2 capture	0.010	0.858	0.700	0.982	0.637
C3 pursuit	0.038	0.165	0.038	0.038	0.070
C4 promotion	0.854	0.809	0.868	0.870	0.850
Mean	0.430	0.654	0.477	0.677	

FIRST DRAFT VS. BEST GAME (MEAN OF 4 GAMES)



FITNESS OF THE BEST GAME SO FAR, PER ITERATION



Conclusions

- The loop helps the improvement: 12 of 16 runs improved (mean fitness 0.26 → 0.56).
- The game matters: balanced games on C4 (0.85), none on C3 (0.07).
- Claude-Sonnet-5 and Qwen3.6-Plus were the strongest and most consistent; GPT-5 and DeepSeek-V4-Pro varied more.

Limitations & next steps

- Game and prompt detail are confounded: each game had its own level of detail.
- One run per model–game pair; no ablation of the loop or its agents yet.
- MCTS may be a weak tester and the set of metrics is limited; no human playtesting.
- Next: the same game at several detail levels, repeated runs, ablations, playtests.

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Scan the QR code for the paper and the code. More results and all details are in the paper.

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